

The Zero That Holds a Place

Answer Key

Grade 2 Math

Quick Answers

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|---|--|
| 1. $500 + 7$ | 6. $600 + 40$ |
| 2. 603 | 7. comparison = $>$, — |
| 3. 209 | 8. (a) the number = 503, (b) expanded form = |
| 4. corrected expanded form = $800 + 30$, — | $500 + 3$ |
| 5. corrected expanded form = $400 + 2$, — | |
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What is verified

5 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 4, 5, 7. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 2 Math

2.NBT.A – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–4 of 5 across 12 tagged checks.

Common wrong answers

1. If they got 57: read the 5 as tens because the tens place is empty
 2. If they got 63: pushed the digits together and dropped the zero placeholder
 3. If they got 29: dropped the zero placeholder and wrote a two-digit number
 4. If they got 803: treated the 3 in the tens place as 3 ones
 5. If they got 420: moved the 2 into the empty tens place
 6. If they got 604: invented a ones part from the tens digit
 8. If they got 53: wrote the two digits side by side with no placeholder
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1. Write 507 in expanded form.

Solution: 507 has 5 hundreds (500), zero tens (write nothing for an empty place), and 7 ones (7). So $507 =$

$500 + 7$

2. Write the number in standard form: $600 + 3$

Solution: 6 hundreds and 3 ones, with *no* tens part — so a zero must hold the tens place between them. Not 63!

603

3. Write the number in standard form: $200 + 9$

Solution: 2 hundreds and 9 ones. The tens place is empty, so a zero holds it: 2 hundreds, 0 tens, 9 ones. 209

4. Find and fix: Nia wrote $830 = 800 + 3$.

Solution: (a) In 830 the 8 counts hundreds and the 3 sits in the *tens* place, so it is worth 30, and the 0 shows there are no ones: $830 =$ $800 + 30$

(b) *The error:* Nia treated the 3 as 3 ones. Her sum $800 + 3$ only makes 803, not 830 — the 3 of 830 is in the tens place and is worth 30.

5. Find and fix: Owen wrote $402 = 400 + 20$.

Solution: (a) In 402 the 2 is in the *ones* place, and the zero in the tens place means no tens part is written: $402 =$ $400 + 2$

(b) *What went wrong:* Owen moved the 2 into the empty tens place. His sum $400 + 20$ makes 420, a different number — the zero of 402 already holds the tens place, so the 2 stays in the ones.

6. Write 640 in expanded form.

Solution: This time the *ones* place holds the zero: 640 has 6 hundreds (600), 4 tens (40), and no ones. So $640 =$ $600 + 40$

7. Kai says 506 and 56 must be equal. True comparison, and why the zero matters.

Solution: (a) 506 has 5 *hundreds*; 56 has only 5 *tens*. A number with hundreds beats a two-digit number: $506 > 56$

(b) *Why the zero matters:* the zero in 506 holds the tens place open, pushing the 5 into the hundreds place. In 56 the 5 only counts tens. Same digits, different places — very different sizes.

8. Challenge: 5 hundreds, no tens, 3 ones.

Solution: (a) Five hundreds, a zero to hold the empty tens place, then 3 ones: 503

(b) Only the places that hold something are written: 500 and 3, so the expanded form is $500 + 3$